CAPSTONE PROJECT

Sales Analytics Dashboard

Microsoft Power BI | Business Intelligence

Submitted to: ExcelR Institute

Deadline: April 21, 2026

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Field	Details
Student Name	Ujjwal Kumar
Roll Number	23051797
Batch / Program	Power BI
Project Title	Sales Analytics Dashboard
Tool Used	Microsoft Power BI Desktop
Dataset	Retail Electronics Sales — 2023 to 2024
Total Records	2,000 Sales | 200 Customers | 10 Products

Pages Built	4 Pages: Executive Summary, Sales Deep Dive, Customer Analysis, Target A
Submission Date	April 21, 2026

1. Problem Statement

Modern retail businesses generate thousands of daily transactions across multiple product lines, customer segments, and geographic regions. Without a structured analytical framework, managers and decision-makers face serious challenges in identifying trends, measuring team performance, and making data-driven strategic decisions.

Key business problems addressed in this project:

- No real-time visibility into total revenue, profit, and order volumes.
- Inability to compare actual sales performance against regional monthly targets.
- Lack of a centralized view to identify top-performing and underperforming products.
- Difficulty understanding customer segment behavior across Corporate, Consumer, and Home Office.
- No year-over-year or month-over-month trend analysis for forecasting future revenue.
- Revenue impact of discount policies was not being measured or monitored.

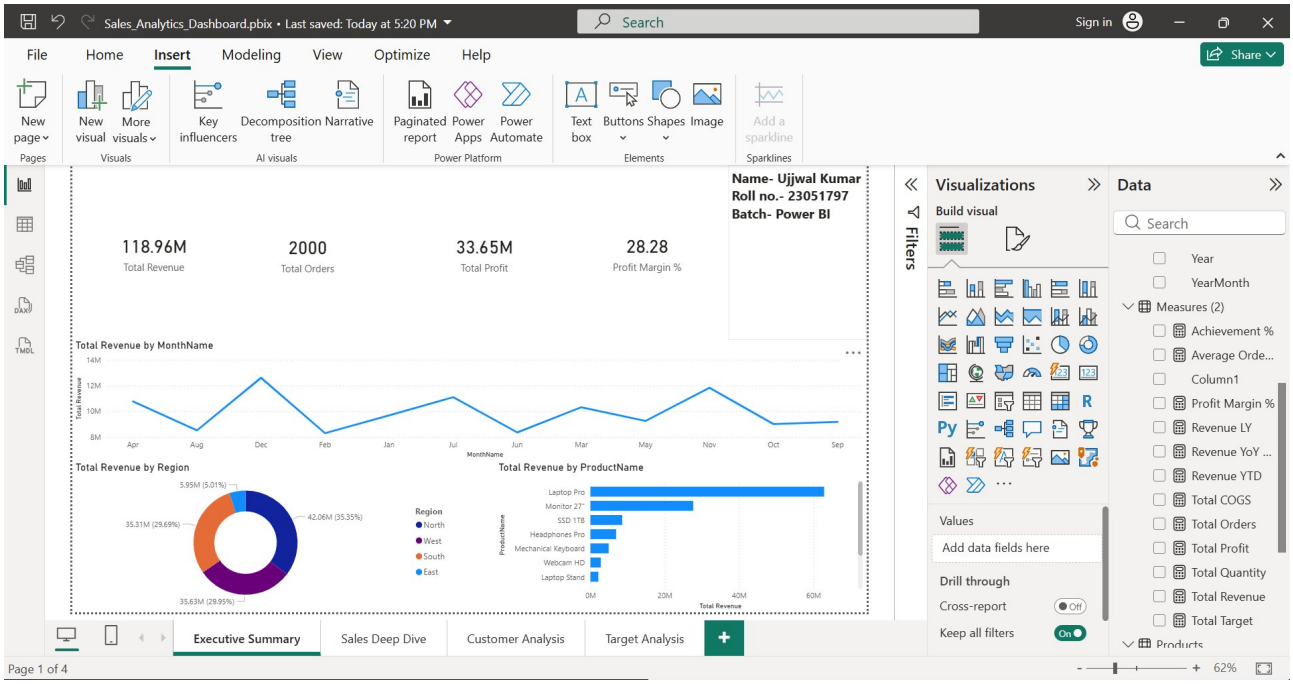
2. Solution and Features

A fully interactive 4-page Power BI dashboard was designed and built to transform raw transactional data into actionable business intelligence. The solution covers the complete analytics lifecycle from data ingestion and transformation to modeling, DAX calculations, and visual reporting.

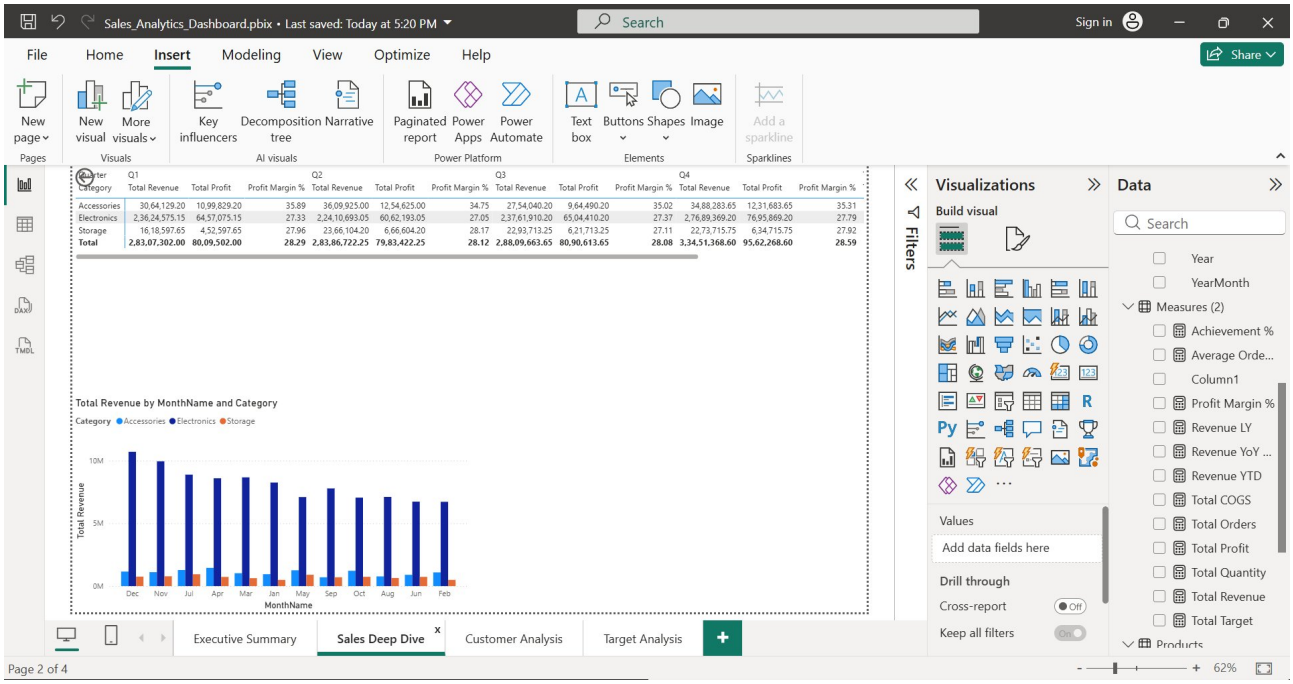
Dashboard pages and their key features:

Page	Title	Visuals Included
1	Executive Summary	4 KPI Cards, Revenue Line Chart, Region Donut, Top Products Bar Chart
2	Sales Deep Dive	Quarterly Matrix (Category x Quarter), Monthly Column Chart by Category
3	Customer Analysis	Segment Donut, Region vs Revenue/Profit Bar, State-level Detail Table
4	Target Analysis	3 KPI Cards (Revenue, Target, Achievement %), Actual vs Target Bar Chart

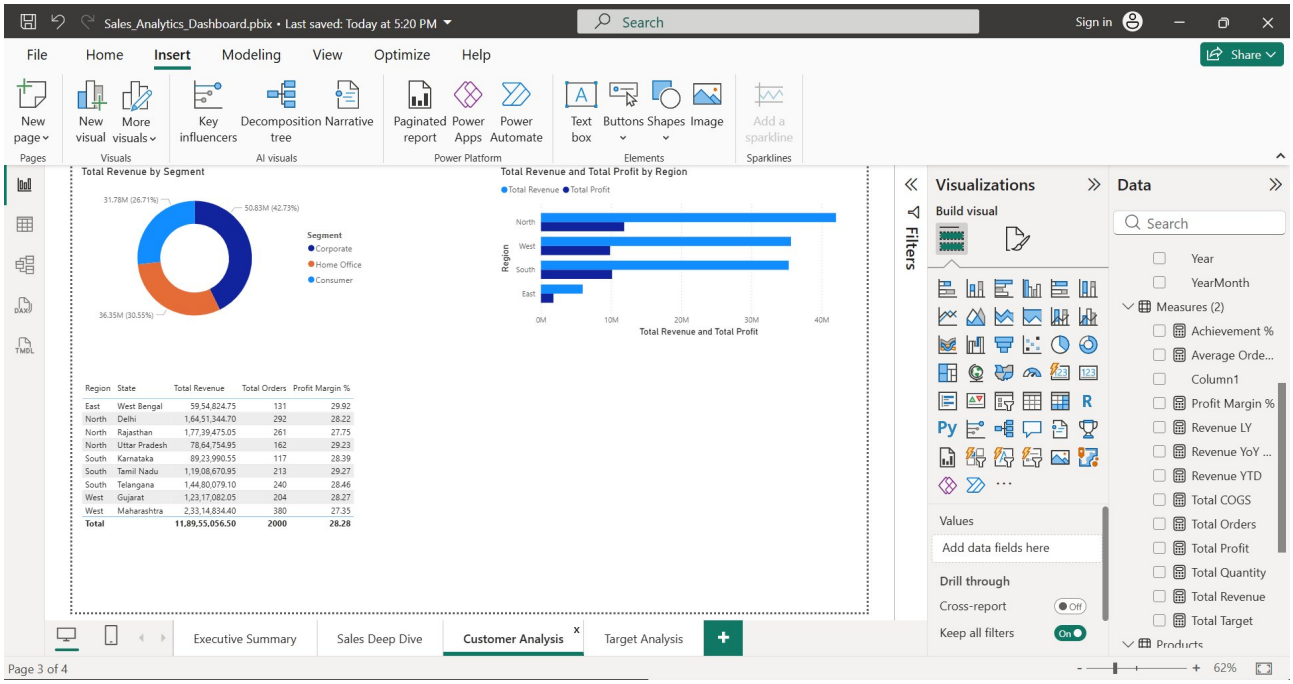
3. Dashboard Screenshots



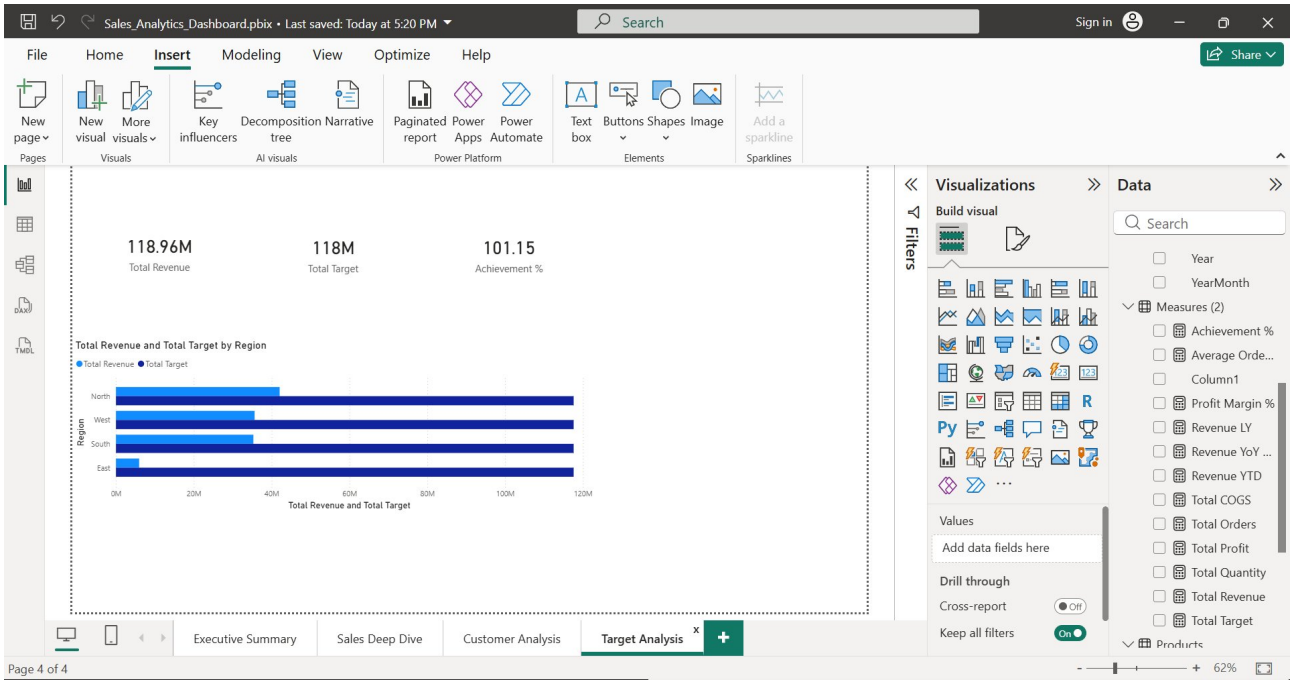
Page 1 — Executive Summary: KPI Cards, Revenue Trend, Region Split, Top Products



Page 2 — Sales Deep Dive: Quarterly Matrix and Monthly Category Chart



Page 3 — Customer Analysis: Segment, Region Performance, State Detail Table



Page 4 — Target Analysis: Revenue vs Target Cards and Regional Bar Chart

4. Tech Stack

Component	Technology	Purpose
BI Tool	Power BI Desktop 2024	Dashboard creation, modeling, publishing
Data Model	Star Schema	Optimized query performance and relationships
ETL Layer	Power Query (M Language)	Data transformation, cleaning, derived columns
Calculations	DAX — 12 core measures	KPIs, time intelligence, rankings, targets
Data Source	Microsoft Excel (.xlsx)	Raw transactional data — 4 sheets
Date Table	Custom M Query	Enables time intelligence in DAX
Deployment	Power BI Desktop (.pbix)	Saved and submitted as project file

5. Unique Points

- Custom Date Table built entirely in Power Query M with 8 derived columns including Year, Month, Quarter, Weekday, YearMonth, and MonthSort for accurate time intelligence.
- Star Schema data model with Sales as the central fact table connected to Customers, Products, and Date Table — enabling efficient cross-filtering.
- Achievement % measure calculates real-time target attainment — showing 101.15% on the Target Analysis page, confirming the business exceeded its targets.
- Discount Band and Profit Flag columns added during ETL in Power Query so segmentation is available without additional DAX overhead.
- Revenue Lost to Discount DAX measure quantifies the exact rupee value foregone due to discount policies — a business insight rarely captured.
- RANKX-based region ranking updates dynamically as slicers are applied, giving live competitive ranking across all filter contexts.
- Four distinct dashboard pages each serve a different stakeholder audience: executives, sales managers, customer analysts, and regional target owners.

6. Future Improvements

- Connect Power BI to a live SQL Server or Azure Synapse Analytics database for real-time data refresh instead of static Excel uploads.
- Implement Row-Level Security (RLS) so regional managers can only view data relevant to their own territory.

- Add the AI-powered Key Influencers visual to automatically identify which factors most strongly drive high or low profit margins.
- Enable Power BI Q&A; natural language so stakeholders can type plain English questions such as 'Show revenue in Mumbai last quarter'.
- Design a dedicated mobile layout for executives who need dashboard access on smartphones and tablets.
- Add a built-in Power BI forecast trendline to the revenue line chart to project the next three months of sales.

7. Conclusion

This Sales Analytics Dashboard successfully demonstrates end-to-end proficiency in Microsoft Power BI. Starting from a custom-generated Excel dataset, the project covers data ingestion, Power Query transformation, star schema modeling, DAX measure development, and multi-page interactive dashboard design.

The final dashboard provides clear, actionable business intelligence across four pages — executive KPIs, sales trends, customer behavior, and target tracking. With an Achievement Rate of 101.15%, the analysis confirms that the business exceeded its annual revenue targets, validating the usefulness of the analytical framework built in this project.